21-242 Recitation 12

RREF — Q6. Row-equivalent matrices have the same RREF

Course-level solution, with a later-course alternative when useful

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Prompt

True/false: If $A = GB$ for some invertible G, then $\mathrm{RREF}(A) = \mathrm{RREF}(B)$.

Course-level solution

True. $A = GB$ with G invertible means A and B are row-equivalent: they have the same row space and, more to the point, each is obtained from the other by elementary row operations. The reduced row echelon form of a matrix is *unique*, so there is only one RREF in a row-equivalence class. Hence $\mathrm{RREF}(A) = \mathrm{RREF}(B)$.
(If uniqueness of RREF has not been proved yet: $\mathrm{RREF}(A) = G_A A = G_A GB$ and $\mathrm{RREF}(B) = G_B B$ are both RREF matrices that are row-equivalent, and one checks that a matrix in RREF is determined by its pivot pattern and row space, or invokes uniqueness.)

Later-course solution (using ideas from the rest of the handout)

Uniqueness of RREF is the key lemma. A quick uniqueness argument: if R and S are both RREF and $R = HS$ for invertible H, then R and S have the same nullspace, hence the same pivot columns (the non-free coordinates), and comparing pivot columns — which are standard basis vectors — forces $H = I$ on the pivot rows, hence $R = S$.